

Problem Setup



Mission context

Earth observation

Plan an Earth observation mission under constraints - here are the handoff docs.

Agent Workspace

- Case package: mission.yaml, satellites.yaml, targets.yaml
- Mission brief: targets, horizon, orbit
- Local workspace: case files, skills, documentation, verifier

Toolbox

- Scientific: numpy, pandas
- Astrodynamics: brahe, poliastro
- Optimization: ortools, pulp

Environment

- Code execution
- Local data and ephemerides
- Outputs and artifacts

Agent Planning



LLM Agent

- Read handoff docs and case files
- Reason over constraints and trade-offs
- Write / adapt code and execute
- Construct executable plan artifact

Physical Feasibility

Visibility windows
Slew / timing limits
Energy and storage

Mission Value

Target priority
Coverage / revisit
Resource efficiency

Planning Artifacts

Illustrative schedule

Time	Asset	Target	Action
Apr 14 09:20	SAT-1	T-07	Image
Apr 14 10:05	SAT-2	GS-Madrid	Downlink
Apr 15 13:40	SAT-1	T-03	Image

External Evaluation



Independent verifier

Checks the submitted plan against the task contract.

Reports validity and mission value.

Evaluation Checklist



Submission received



Schema valid



Timing feasible



Geometry feasible



Resources within limits



Results

Family-native mission metrics completion, coverage, efficiency



Validity



Normalized score